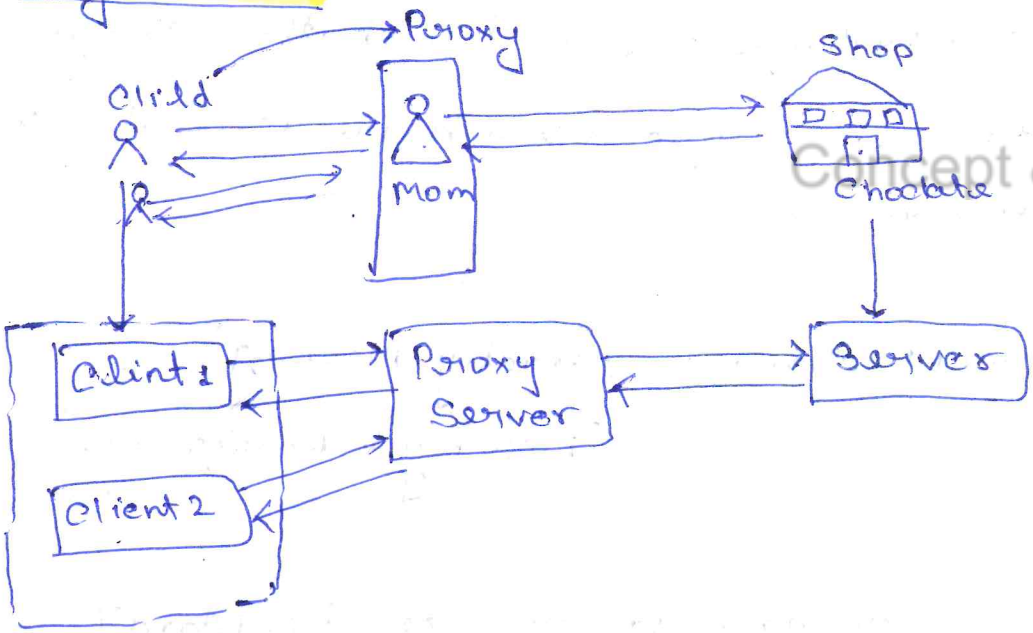
17. Proxy vs Reverse Proxy | How Proxy differs from VPN, Load Balancer

Lecture - 54

- What is Proxy Server
- What are its different types?
  - 1 Forward Proxy
  - 2 Reverse Proxy
- Proxy vs VPN
- Proxy vs Load Balancer
- Proxy vs Firewall.

Proxy Server



Concept && Coding

Proxy Server : Proxy Server sit between client and Server. It's act like a intermediate. between client and Server. Request can pass to proxy. No direct interaction

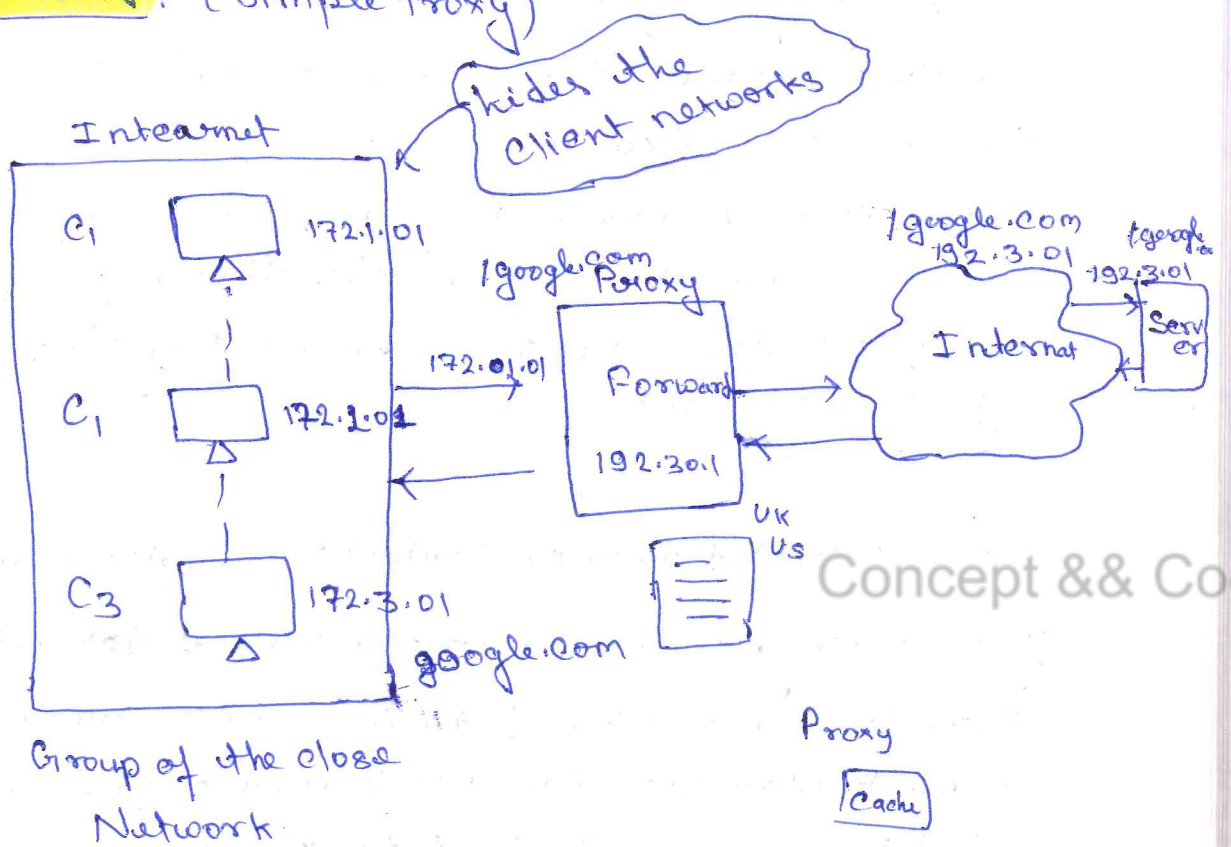
②

Types of Proxy

(differ in direction of Communication)

- Forward
- Reverse

1. **Forward** :- (Simple Proxy)



Concept && Coding

What does forward proxy do.??

Ans.: It's hides the client networks, means server Outside Server don't know Internal client.

* How??

Let say Client one makes a request so its IP address 172.1.01. One makes a request. now request is comming from C1 then Proxy will sends its own IP address to the outer word like this is making a request. on behalf of C1. So Server when it gets a request hey

Advantages:-

① It's gives anonymity to the internal Networks: so It's IP address, It's Location it's got hidden to the outside wall no server can know that request comes from where.

② Grouping of request. A proxy can take care of multiple ~~requ~~ client requests. Let's say this client is also making a google.com request this client is also making google.com request so what it can do it can club it can group the request and send one request say send the google.com outside.

3:- Access Restricted data or content:- So let's say your country has blocked certaint kind of ~~dat~~ Contents that it is not allowed to visit. let's say in India a particular websites are banned or you cannot visit those. so with help of proxy you can bypass those because you can this might be you let's say it will proxy as like this request is comming from let's say some UK, US. ~~a~~ User in India and it is trying to access certain restrictive content. because of country wise location restrictive those content can't access.

Concept && Coding

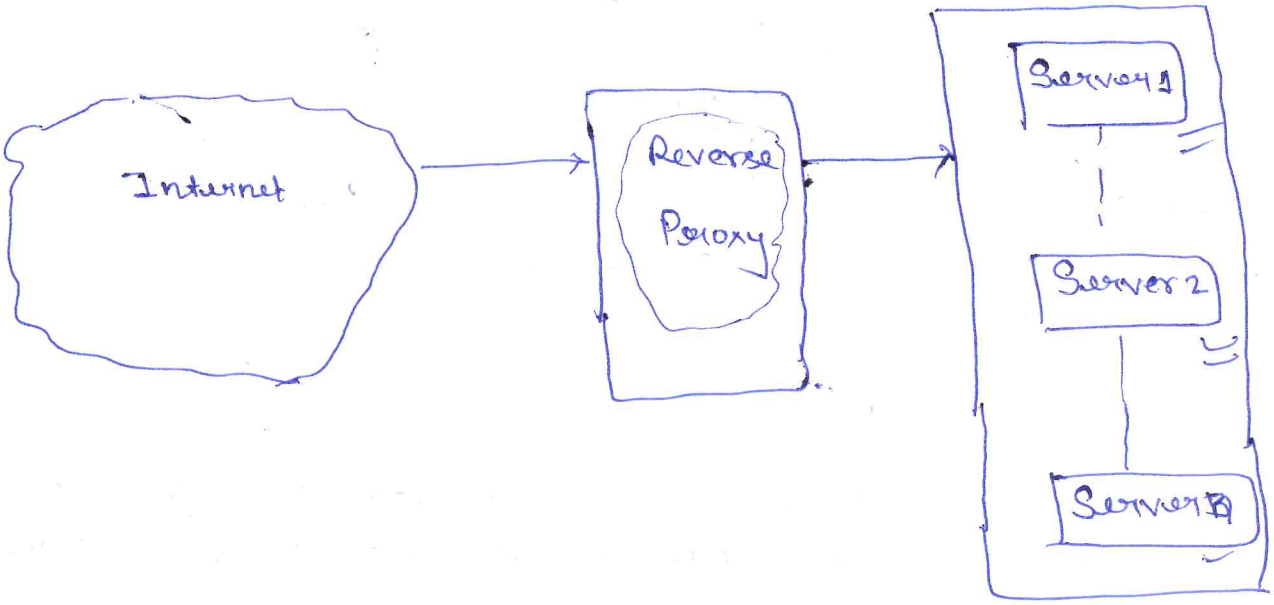
4:- It's Bring Security :-

5 Caching :-

Disadvantages

This work on application Level.

2 :- Reverse Proxy :-



- A Reverse Proxy is like a Security gaurd for your backend servers. Instead of users directly talking to your servers, they talk to the reverse proxy, which them forwards the request to the appropriate server

Example :- Imagin a resturnent with waiter (reverse proxy). Instead of customers going directly to the Kitchen (backen servers), they order through the waiter. The waiter then communicates with the Kitchen and brings back the food.

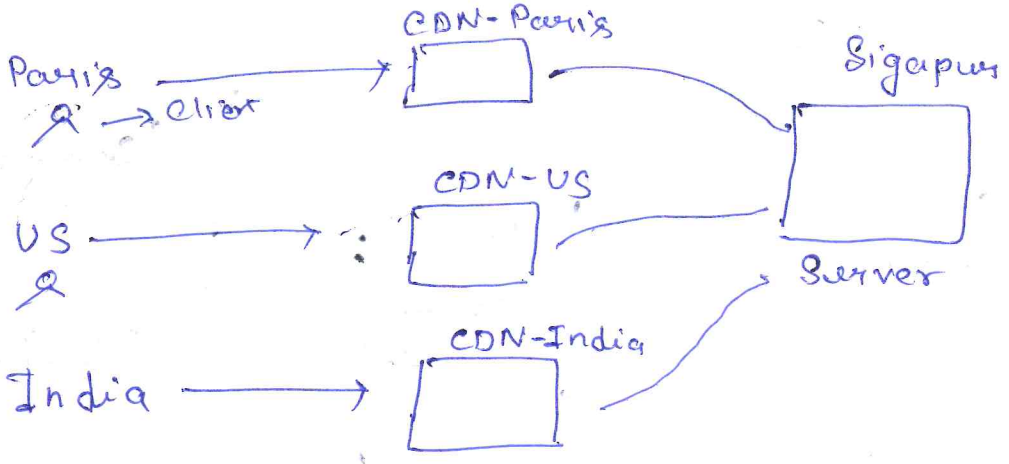
Concept && Coding

Advantages :-

⇒ Security :- Outside world don't aware of Servers Ip address. All aware of Reverse Proxy address.

In case of DDOS Attacked :- Attacker can only attack reverse proxy ~~Server~~.

CDN ← Reverse Proxy



Local client when their request forms it goes to their local CDN. Attacker can attack only a CDN.

(2) Also have capable to caching.

(3) Lactency :-

(5) Load Balancer capbility :-

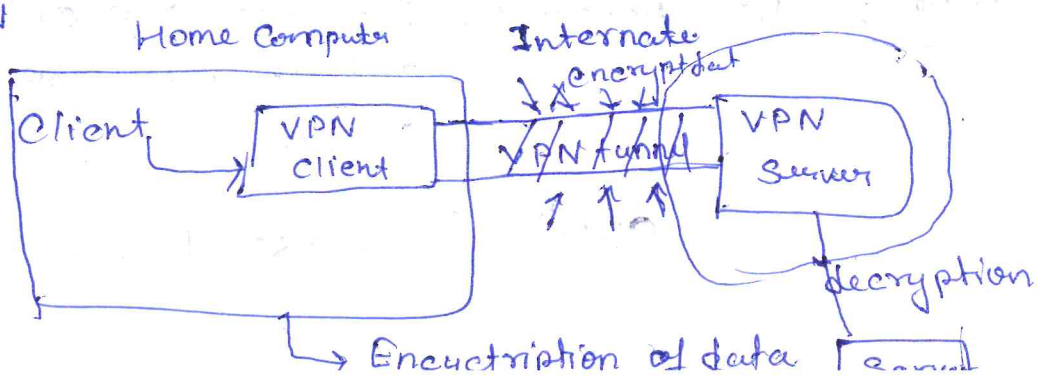
Concept && Coding

Proxy VS VPN

Client → Proxy → Server

anonymity (IP address) masking
Caching
Logging

VPN

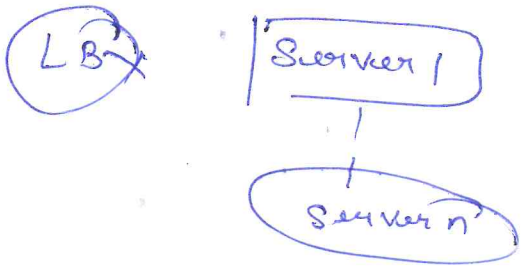


Reverse Proxy vs Load Balancer

IP anonomity
caching
Logging

→ R. Proxy act as a Load Balancer

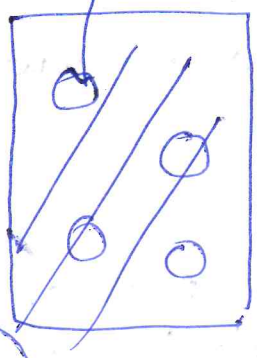
→ Load Balancer can not be a proxy



Proxy vs FireWall

Packet Scanning

rule : What are data pass in the hole.



Header
Port No
IP Address
Source/destination

Internet

Concept && Coding